

TASK 9

1.Source code

1.Student Class – Student.java

```
import java.util.Objects;
```

```
/*
```

```
 * Student model class
```

```
*/
```

```
public class Student {
```

```
    private int id;
```

```
    private String name;
```

```
    private int marks;
```

```
    public Student(int id, String name, int marks) {
```

```
        this.id = id;
```

```
        this.name = name;
```

```
        this.marks = marks;
```

```
    }
```

```
    public int getId() {
```

```
        return id;
```

```
    }
```

```
    public String getName() {
```

```
        return name;
```

```
}
```

```
public int getMarks() {
```

```
    return marks;
```

```
}
```

```
@Override
```

```
public String toString() {
```

```
    return String.format("ID: %d | Name: %s | Marks: %d", id, name, marks);
```

```
}
```

```
// Used for removing duplicates in Set
```

```
@Override
```

```
public boolean equals(Object o) {
```

```
    if (this == o) return true;
```

```
    if (!(o instanceof Student)) return false;
```

```
    Student student = (Student) o;
```

```
    return id == student.id;
```

```
}
```

```
@Override
```

```
public int hashCode() {
```

```
    return Objects.hash(id);
```

```
}
```

```
}
```

2.Student Manager – StudentManager.java

```
import java.util.*;

/*
 * Handles collection operations
 */

public class StudentManager {

    // ArrayList to store students
    private List<Student> studentList = new ArrayList<>();

    // HashMap for fast lookup
    private Map<Integer, Student> studentMap = new HashMap<>();

    // Add student
    public void addStudent(Student student) {
        studentList.add(student);
        studentMap.put(student.getId(), student);
    }

    // Remove duplicates using Set
    public void removeDuplicates() {
        Set<Student> uniqueStudents = new HashSet<>(studentList);
        studentList.clear();
        studentList.addAll(uniqueStudents);
    }
}
```

```

    }

    // Sort students by marks using Comparator
    public void sortByMarks() {
        studentList.sort(Comparator.comparingInt(Student::getMarks).reversed());
    }

    // Get student by ID (HashMap lookup)
    public Student getStudentById(int id) {
        return studentMap.get(id);
    }

    // Display report
    public void printReport() {
        System.out.println("\n--- Student Report ---");
        for (Student s : studentList) {
            System.out.println(s);
        }
    }
}

```

3.Main Application – StudentApp.java

```

/*
 * Main application class
 */

```

```
public class StudentApp {  
  
    public static void main(String[] args) {  
  
        StudentManager manager = new StudentManager();  
  
        // Adding students (including duplicate)  
        manager.addStudent(new Student(1, "Akshu", 85));  
        manager.addStudent(new Student(2, "Riya", 92));  
        manager.addStudent(new Student(3, "Anu", 78));  
        manager.addStudent(new Student(1, "Akshu", 85)); // duplicate  
  
        // Remove duplicates  
        manager.removeDuplicates();  
  
        // Sort by marks  
        manager.sortByMarks();  
  
        // Print report  
        manager.printReport();  
  
        // HashMap fast lookup  
        System.out.println("\nSearching for student with ID 2:");  
        System.out.println(manager.getStudentById(2));  
    }  
}
```

```
}
```

2.Output

```
--- Student Report ---  
ID: 2 | Name: Riya | Marks: 92  
ID: 1 | Name: Akshu | Marks: 85  
ID: 3 | Name: Anu | Marks: 78  
  
Searching for student with ID 2:  
ID: 2 | Name: Riya | Marks: 92
```